Chapter 21 | Complete worked solutions

Image and Video Understanding

All twelve exercises are solved here. Begin with shapes, preprocessing and attention counts, then trace real array values, train small linear probes, and design evaluation protocols. The examples run on CPU without downloads. The image features and numeric fixtures are deliberately small; they are not benchmarks of pretrained vision models. The written PDF preserves every explanation and both plots; the notebook PDF also includes code and outputs.

1. Make a patch grid

The image has 12 rows and 8 columns. A patch spans four rows and four columns, so there are $12/4 = 3$ patch rows and $8/4 = 2$ patch columns. This gives six patches. Each contains $4 \times 4 \times 3 = 48$ channel values.

Flattening only rearranges those values into a row. Stacking the six rows gives shape $[6, 48]$. Multiplying by a shared projection of shape $[48, 5]$ gives $[6, 5]$. One classification row gives the final shape $[7, 5]$; its position array must also have shape $[7, 5]$. No batch axis is shown.

A linear projection from 48 dimensions to five cannot distinguish every possible real-valued patch: different inputs can map to the same output. Flattening does not have this limitation because it can be reversed exactly when the original dimensions and order are known. The classification row is a learned summary position, not another observed image patch.

2. Change the patch size

For a 32×32 image, patches of side eight give a 4×4 grid, so $N = 16$ and the full patch-only score matrix has $16^2 = 256$ entries. Patches of side four give an 8×8 grid, so $N = 64$ and $64^2 = 4096$. The token count grows fourfold; the score count grows sixteenfold.

A smaller patch lets attention address smaller regions separately. Two objects that previously fell inside one token can occupy different tokens. This does not guarantee a better classifier. Pixel evidence, the projection, training, and task all matter. Nor is this a sixteenfold claim about total runtime: projections and feed-forward layers have other costs. Adding a classification token would give $17^2 = 289$ and $65^2 = 4225$ scores, so the exact sixteenfold ratio applies only to the patch-only count requested.

3. Check an input convention

First match the pixel scale: $153/255 = 0.6$. Then apply the channel's normalization: $(0.6 - 0.5)/0.25 = 0.4$. Skipping the first operation gives $(153 - 0.5)/0.25 = 610$, an entirely different input.

The mean and standard deviation are meaningful on their specified scale. Use the checkpoint's own preprocessing; the example statistics are not universal. Two other choices to match are channel order, such as RGB rather than BGR, and the resize/crop procedure including its target dimensions. A correct numerical range cannot restore an object that a crop removed. Interpolation and padding conventions can matter as well.

4. Count video connections

There are $T = 4$ frames and $S = 9$ patches per frame, so the clip has $TS = 36$ patch tokens. Joint attention has $36^2 = 1296$ ordered query-key scores per head. Temporal attention has nine groups of four tokens, $9 \times 4^2 = 144$ scores. Spatial attention has four groups of nine tokens, $4 \times 9^2 = 324$ scores. Their sum is 468. These counts include self-pairs in each operation and exclude special tokens.

One temporal group is $(1, 2)$, $(2, 2)$, $(3, 2)$, $(4, 2)$, with frame index first. One spatial group is $(4, 1)$ through $(4, 9)$. The desired route is input $(1, 2)$ to the temporally mixed output at $(4, 2)$, then to the spatially mixed output at $(4, 7)$. Each arrow carries information from a value source into a receiving output. The intermediate token belongs to a new stage, not to the original input array. This possible route explains reachability; it does not make two operations equal to joint attention or guarantee a large learned weight on that route.

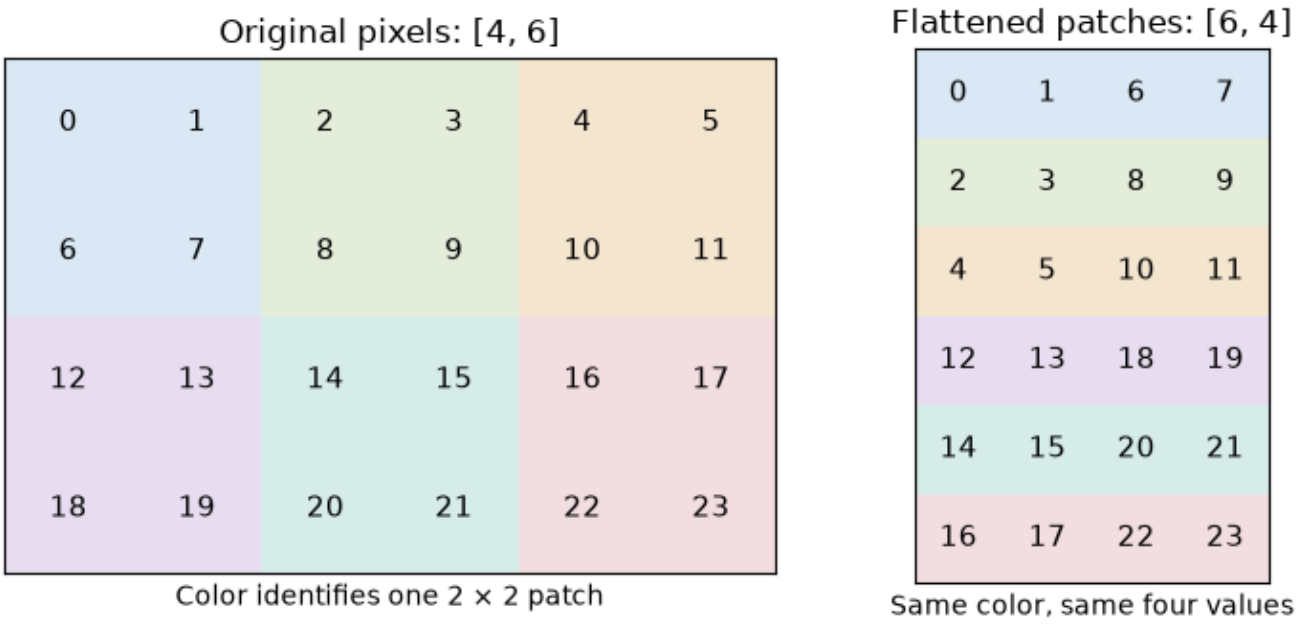
5. Check row-major patch order

Fill a four-row, six-column image with the integers 0 through 23. Each pixel has one channel. Think of the values as labels for the pixels. They let us trace each entry in a flattened patch back to its original location. A side length of two gives two patch rows and three patch columns. Each patch contributes four values, so the output should have shape $(6, 4)$.

Row-major means `patchify` visits the three patches across the top before moving to the bottom. It also reads the pixels within each patch across the first row, then across the second. The first output row comes from the upper-left patch, and the last comes from the lower-right patch. With zero-based image coordinates and the single channel fixed at 0, the entries map back as follows.

Output row	Pixel value	Image row	Image column
First	0	0	0
First	1	0	1
First	6	1	0
First	7	1	1
Last	16	2	4
Last	17	2	5
Last	22	3	4
Last	23	3	5

The value at image row r and column c is $6r + c$ because each preceding image row contains six pixels. To check the ordering independently, we can read all six patches directly by their upper-left corners and compare the four entries in each row. A shape check alone would also pass if the patch rows were shuffled.



Each colored region on the left becomes one row on the right. The first row is 0, 1, 6, 7 and the last is 16, 17, 22, 23. The figure traces values, rather than merely checking that both arrays contain 24 entries.

6. Add a position to an identical patch

Give two patches the same four input values, $(1, 2, 3, 4)$. A projection turns each length-four row into a length-two token by multiplying it by the same weight matrix. We choose the first column of the matrix to add the first and third values, and the second column to subtract the fourth value from the second.

$$(1, 2, 3, 4) \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} = (1 + 3,\ 2 - 4) = (4, -2).$$

Each output coordinate is a dot product with one column of the matrix. The two input rows are identical, so both dot products give the same results for each row. Call `patch_embeddings` without a classification token. Both calls therefore return a $(2, 2)$ array, with one row per patch and one column per output coordinate.

`positions` supplies an additive vector for each token row. Adding $(0, 0)$ leaves the first row at $(4, -2)$. Adding $(0, 1)$ changes the second row to $(4, -1)$.

More generally, let z be the shared projected patch and p_1 and p_2 be any two position vectors of the same width. The returned rows are $z + p_1$ and $z + p_2$. Subtracting the first row from the second cancels z and gives $p_2 - p_1$. Adding the same shift to both positions moves both tokens by that amount and preserves their difference. The exercise chooses the two vectors by hand to make this operation easy to see. In the chapter's Vision Transformer, training learns a position vector for each location.

The rows differ because different position vectors were added to the same content vector. Simply placing identical vectors in two rows would not make them different. More generally, in deterministic full self-attention with shared tokenwise operations and no position-dependent mechanism, permuting patch rows permutes their outputs. A classification token kept in place has the same output. Fixed location vectors break this symmetry when content moves relative to them. Permuting content together with its position vectors only relabels the rows again. A mask or other mechanism that depends on position changes the assumptions of this argument.

7. Compare ordered and sorted features

The four features give the brightness at the upper left, upper right, lower left, and lower right, in that order. Class 0 has its bright patch in the top row and class 1 has it in the bottom row. For example, $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ place the same brightness in different rows. An ordered head can separate the classes by comparing the sum of the two upper features with the sum of the two lower ones.

Fit a **linear probe**, a classifier that learns weights and biases while keeping the input features fixed. Both fits use four input features, two class scores, 120 training steps and a learning rate of 0.5. The second fit sorts values within each row, preserving which brightness values appear while discarding their locations. The labels stay attached to the same training examples.

Multiply the feature rows by the fitted weight matrix and add the bias to get two raw scores, or *logits*, for each example. Column 0 scores class 0 and column 1 scores class 1. Choose the larger score on each row. When two scores tie, NumPy's `argmax` returns the first index, which here is class 0.

The ordered head predicts $(0, 1, 0, 1)$ and gets all four rows right. In the sorted fit, each distinct input occurs once with each label. The head starts with zero weights and biases, so both classes start with probability $1/2$. The two examples in each pair give equal and opposite updates because their features are identical and their labels differ. The updates cancel at every step, leaving scores $(0, 0)$ on all four rows. The tie rule gives predictions $(0, 0, 0, 0)$ and two correct answers.

The pairwise limit holds for **any deterministic classifier** using only these sorted inputs. A brightness of 1 can still be distinguished from a brightness of 2, but both examples with the same brightness must receive the same prediction. Each pair contains one example of each class, so exactly one prediction per pair is correct regardless of the chosen class.

These are predictions on the four rows used for fitting. The chapter's separate synthetic-image experiment evaluates generalization to held-out brightness and background settings. There the ordered head gets 8 of 8 and the sorted head gets 4 of 8.

The four coordinates in this exercise are fixed spatial features read by separate head weights. They are not patch tokens passed through a position-free ViT. Equal feature counts and equal head capacities isolate the information loss caused by sorting. These predictions are on the four training rows; the companion runner separately uses 12 training and eight held-out synthetic images, obtaining 8/8 versus 4/8 on the held-out set.

8. Trade temporal for spatial tokens (optional)

A token has a frame index and a patch location within that frame. Joint attention can score a query token against every key token in the clip. Divided attention first scores tokens at the same patch location across frames, then scores tokens in the same frame across patch locations.

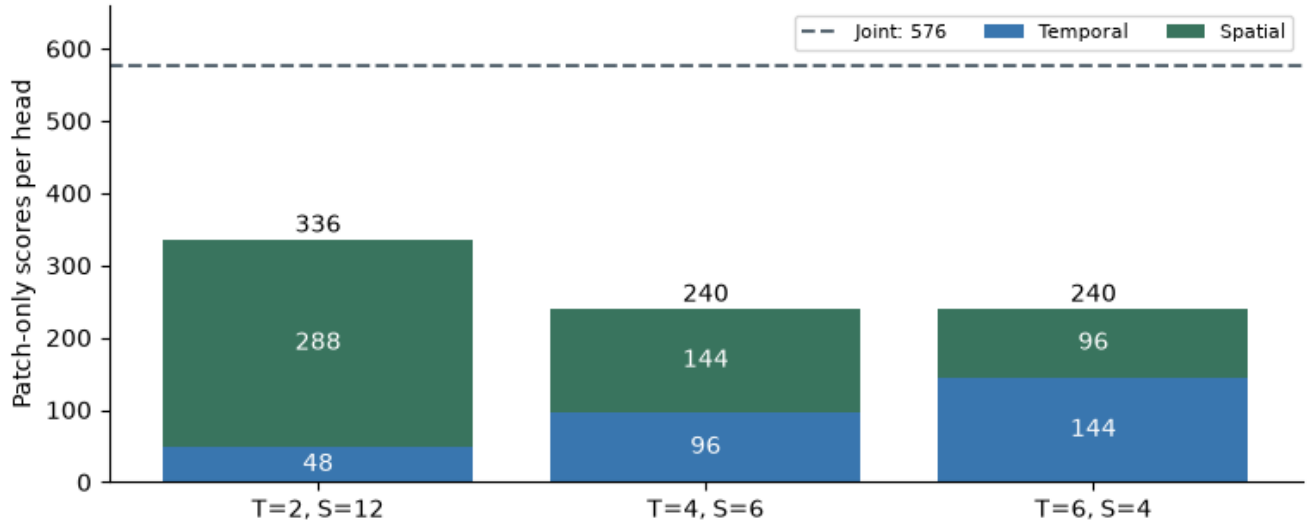
Count one entry per ordered query and key pair, per attention head. Include a token's score against itself and exclude special tokens. A pair from token A to token B and the reverse pair occupy separate entries. A token's self-score appears once in each of the two divided operations.
With T frames and S patches per frame, the clip has TS tokens. There are S temporal groups of size T and T spatial groups of size S, so the counts are

$$\begin{aligned} \text{Joint} &= (TS)^2, \\ \text{Temporal} &= ST^2, \\ \text{Spatial} &= TS^2, \\ \text{Divided} &= ST^2 + TS^2 = TS(T + S). \end{aligned}$$

For example, $(T, S) = (2, 12)$ has 12 temporal groups with 2^2 entries each and two spatial groups with 12^2 entries each. Check the function by explicitly counting the pairs of token coordinates that each operation scores.

Every clip has 24 tokens and $24^2 = 576$ joint scores. The divided count drops from $24(2 + 12) = 336$ to $24(4+6) = 240$ as the group sizes become closer. Swapping 4 and 6 interchanges the temporal and spatial counts, 96 and 144, but their sum stays 240. Both TS and $T + S$ are unchanged by the swap. The two operations still group the tokens along different axes, so the equal totals describe the number of entries each block scores.

Writing $N = TS$, the divided count is $ST^2 + TS^2 = N(T + S)$. Dividing by the joint count N^2 gives $1/T + 1/S$. For the three grids the ratios are $7/12$, $5/12$, and $5/12$. With a fixed product TS, balancing the two dimensions reduces their sum; swapping the dimensions preserves it. A frame and a patch location still represent different things. Equal counts do not imply equal information, predictions, or runtime.



Blue and green portions count the two divided operations, including self-pairs in each. All three clips contain 24 tokens, so their joint count is the same dashed reference. The first divided total is 336; the other two are 240. The bars measure scores, not elapsed time.

9. Form tubelets (optional)

Six frames divided into groups of two give three temporal groups. An 8×12 frame divided into 4×4 patches gives two patch rows and three patch columns. Thus $3 \times 2 \times 3 = 18$ tubelets cover the clip. Each contains $2 \times 4 \times 4 \times 3 = 96$ RGB channel values.

Taking patches separately from each frame gives $6 \times 2 \times 3 = 36$ tokens, each starting with 48 values. The tubelets halve the token count but double the input width of each token's projection. Both constructions account for $18 \times 96 = 36 \times 48 = 1728$ channel values before projection. The tubelet projection sees two frames together; it need not discard all order information, because those frames occupy distinct input coordinates. A narrower output can still lose information. Fewer tokens alone do not establish equal quality or a proportional saving in total compute.

10. Find a leaking split (optional)

Split original recordings before making overlapping clips. For example, assign R1-R7 to training, R8 to validation, and R9-R10 to test. With three clips per recording, these contain 21, 3, and 6 clips. Keep all augmentations, crops and derived clips with their parent recording. Verify that parent IDs are disjoint. Randomly splitting the 30 clips could put nearly identical frames on both sides and reward familiarity with a scene or actor.

For generalization to new cameras, group by camera as well. Suppose camera A supplies R1-R4, B supplies R5-R7, C supplies R8, and D supplies R9-R10. Train on A and B, validate on C, and test on D. This example has the same clip counts, but its unit of exclusion is stronger. New recordings from camera A do not constitute a test on a new camera. If people or locations cross camera groups and must also be unseen, define groups that prevent those overlaps too.

Choose the aggregation rule and tuning decisions using validation only. For example, average clip probabilities within each recording and score one recording prediction. Report class counts and per-class recall. With only two test recordings and one test camera, conclusions about generalization are necessarily limited; more clips do not create more independent cameras. Uncertainty calculations should respect the independent sampling unit. The following fixture checks split membership, not classifier performance.

11. Test an action's direction (optional)

Use opening versus closing a drawer. Select eight frames that visibly cover the action, and pair them with the same frames in reverse order. Verify that reversal really changes the intended label; ambiguous pauses and edits should not be relabeled mechanically. Keep the pair in one recording group when splitting. Include naturally recorded examples of both actions so the experiment does not only distinguish artifacts of reversed footage.

Use the same frozen frame encoder, preprocessing and eight sampled frames for both systems. The baseline averages its per-frame class scores. The comparison model processes frame features with temporal attention and temporal position embeddings. Retrain its head and temporal component on the training split, choose settings on validation, and report their additional parameters and cost. A separate order ablation can remove temporal position information or shuffle frames, but neither alone proves how a trained model uses order.
For any deterministic frame scoring function f,

$$\frac{1}{8}\sum_{t=1}^{8} f(x_t) = \frac{1}{8}\sum_{t=1}^{8} f(x_{9-t}).$$

Thus the averaged-image baseline makes the same prediction for both members. On balanced opposite-label pairs its clip accuracy is exactly one half for a binary forced-choice classifier, and its pair accuracy is zero. Pair accuracy counts a pair correct only when both labels are correct. Report both measures, class recall, and the number of independent recordings. A label flip alone is insufficient: both predictions could have the wrong direction. Use recording-level resampling for uncertainty, keeping each pair together. Temporal attention with position information can represent order, but may still fail to learn it.

The numeric check uses an explicitly supplied two-frame score fixture. It verifies averaging invariance and the scoring rule; it is not a run of either proposed video model.

12. Compare two visual checkpoints (optional)

Use a common supported resolution and identical image crops to compare the checkpoints on the same visible evidence. Keep the dataset split, label mapping, metric, precision and evaluation mode fixed. Record each checkpoint's patch size, model width, depth and pretraining. This comparison answers which complete checkpoint works better under that shared input protocol. It does not isolate patch size: training data, architecture and optimization may differ too.

For a second experiment, hold one checkpoint's weights fixed and vary resolution using its supported position-grid interpolation and preprocessing. Use the same underlying crop region at each size. This measures that checkpoint's response to input resolution, including changed sampling of visual detail and token count. It is not an architecture comparison, and interpolated positions alone do not guarantee good performance at a new size.
For an illustrative model A with patch side 16 and model B with side 8, both at resolution 224, and A again at 320, the counts are:

Setting	Patch tokens	With one classification token	Full scores per head
A: 224, patch 16	196	197	38,809
B: 224, patch 8	784	785	616,225
A: 320, patch 16	400	401	160,801

These include the classification token and differ from the patch-only counts in Exercises 2, 4 and 8. Report the actual score grid, parameters, quality, throughput, batch size, and peak memory. Measure latency after warm-up on the same hardware and software; synchronize asynchronous accelerators around timed work. Show a median and a tail statistic such as the 95th percentile over repeated runs. Specify whether decoding and preprocessing are included. Distinguish model-only from end-to-end timing if both are relevant. The table supplies no timing or accuracy claim.